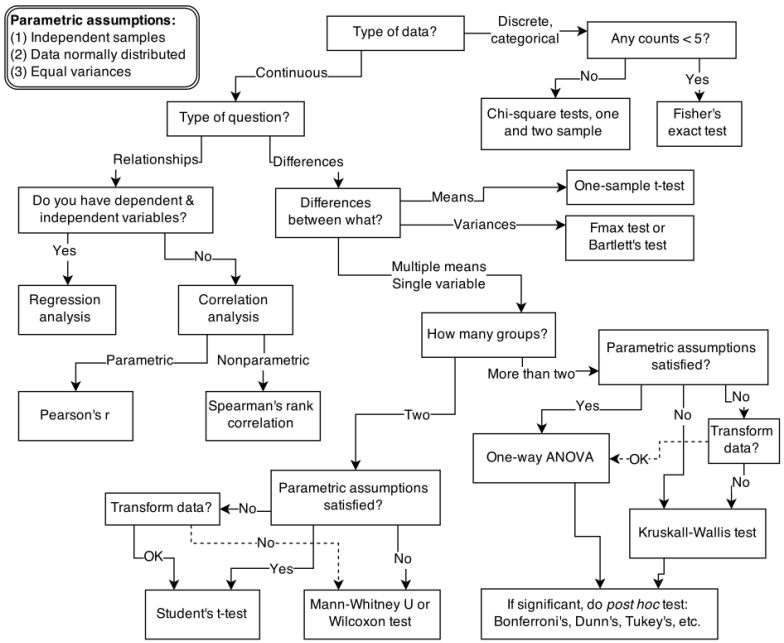
Lecture 3: Chapter 1 - The Golem of Prague

- Statistics for Science course!
- Fascination for 'natural' phenomena like the study of a peculiar species of interest (evolution of a given bird, resistance to disease/yield of a crop, human health...) quickly leads us to turn into data about survival time, energy content, optimal environment (temperature, soil content...), etc.
- That's how we record the characteristics, the variations, the interactions between the components of a (natural) complex system.
- Need maths to describe the world at all scale and make sense of it: large (climate, weather, tectonic plate movements, planets) or small (social sciences).
- How do we analyse these data?

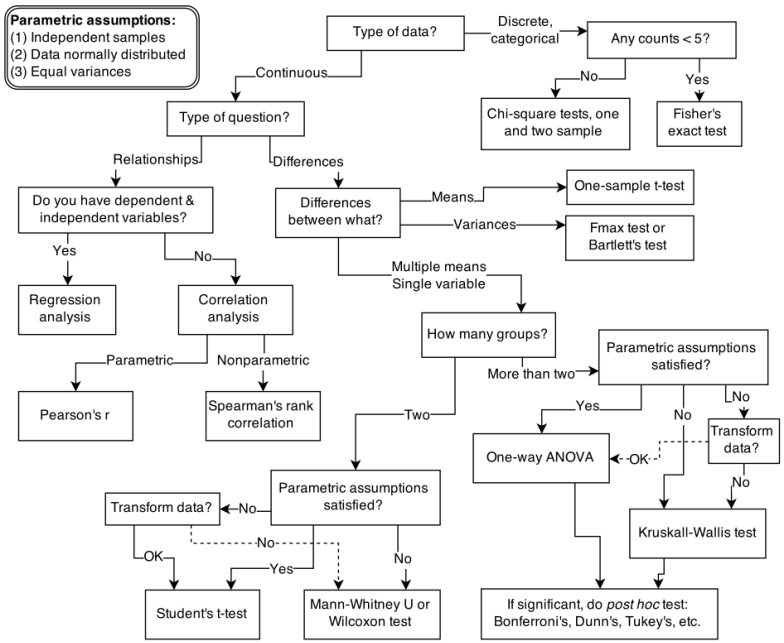
MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Statistical hypothesis testing procedures



(from SR, Chapter 1)

Statistical hypothesis testing procedures



(from SR, Chapter 1)

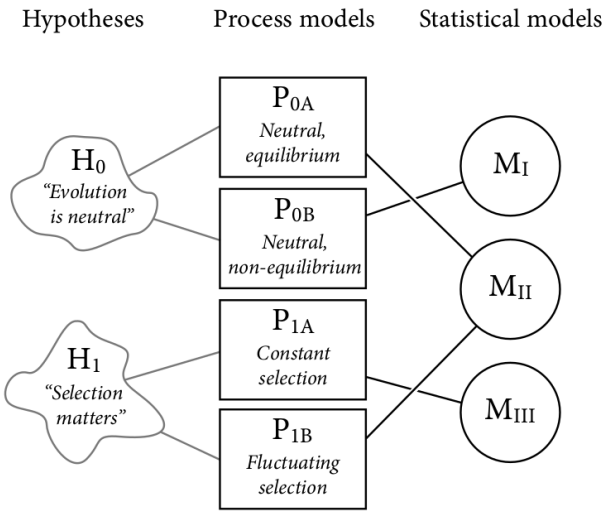
$\rightarrow$ OUCH!! How is the statistical test (and their associated sacred p-value) related to the **research** hypothesis? These constrained tools can be frustrating. Inflexible tools. Where is causation?

MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Golems as scientific models

- A Golem is metaphorically:
  - A clay robot, well possibly a silicon robot living in a computer code
  - They are powerful (no human can do these calculations)
  - But without free will or wisdom; only carry their instructions
  - Hence can be dangerous, when used in the wrong place.
- Golems are just like statistical models for tests from the previous flowchart.
- They can be good in some highly controlled experiment environments, e.g. industrial framework.

Hypotheses, process models and statistical models relationships



(from SR, Chapter 1)

- We want to be careful about the relationships between scientific hypotheses, the scientific process models that emerges form those hypotheses and the statistical models we use to connect these processes to evidence.
- Example: how does DNA evolve?

MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Example from population genetics: how does DNA evolve?

- Follow the H_0 "Evolution is neutral" (mutations explain evolution) is vague $\leftrightarrow$ $P_{0,A}$: no selection (neutral) and constant pop size (equilibrium) $\leftrightarrow$ M_{II}: examine associations.
- Needs both the process and the statistical models. In 20th century, it was solved as $P_{0,A}$ implies a particular distribution of the frequencies of alleles in the population.
- Other processes and statistical models are possible $H_0 \leftrightarrow P_{0,B}$ - pop size can fluctuate; more realistic) $\leftrightarrow$ M_I - different statistical implications. So we can test both against each other.
- A different research hypothesis is possible: H_1 "Selection matters". To what extent does it matter for the frequencies of alleles in the population?
- Issue: $P_{0,A}$ (neutral, equilibrium) and $P_{1,B}$ (selection, non-equilibrium) generate the same distribution...So cannot be told apart.

Limitations of the necessary golems

- The previous scenario happens in many situations: null models are not unique!
- No null phylogeny, no null ecological community, no null social network, no null gene regulatory network, etc.
- We need process models and study the implications of those models.
- Then, statistical models (our golems) need to be justified by these implications and the scientific question, aka the estimand: what we are trying to estimate.

How to draw the owl - step 1



1. Draw some circles

How to draw the owl - step 2

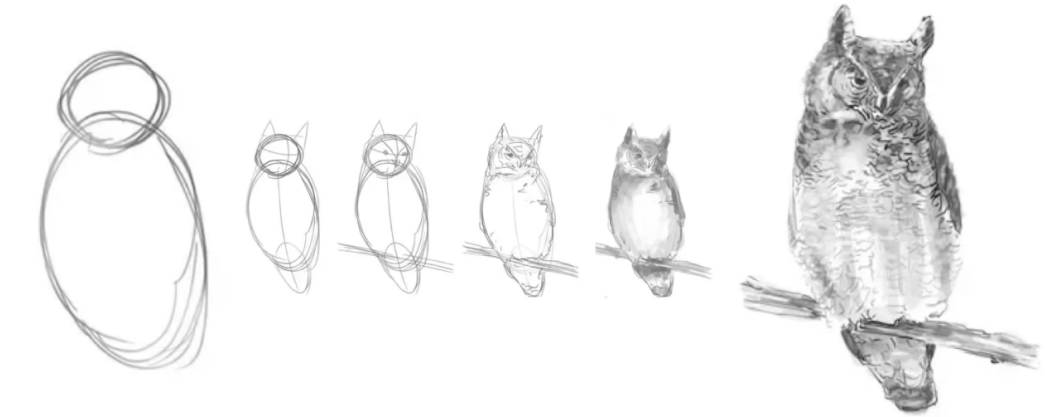


1. Draw some circles



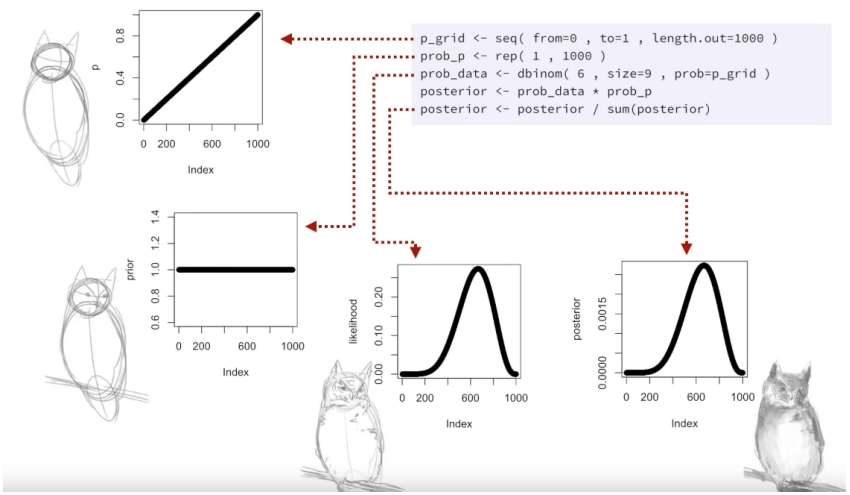
2. Draw the rest of the owl

How to draw the owl: with the missing steps!



- This is more useful.
- Experts are not always aware of all the steps and their expertise.
- We will express statistical/scientific models in code. In details.

Bayesian owl: calculation of a posterior distribution



- No mathematical formula, rather:
- (i) Understand what you are doing (avoid not understanding someone else's black-box) (ii) document your work (future you will thank you), reduce some errors and reuse it and (iii) respect scientific workflow: others can follow it.

To draw the Bayesian awl

You need

1. To understand the (theoretical) estimand: what are you trying to do? Often not that easy to tell what a scientific study is trying to do. No vague research description.

To draw the Bayesian awl

You need

1. To understand the (theoretical) estimand: what are you trying to do? Often not that easy to tell what a scientific study is trying to do. No vague research description.
2. A scientific (causal) model; estimand in 1. built in this context. We may resort to simulated data to convince ourselves.

To draw the Bayesian awl

You need

1. To understand the (theoretical) estimand: what are you trying to do? Often not that easy to tell what a scientific study is trying to do. No vague research description.
2. A scientific (causal) model; estimand in 1. built in this context. We may resort to simulated data to convince ourselves.
3. To build a statistical models using 1. & 2. above. Can get out the estimand. Sometimes we cannot extract the estimand, but at least we will know we need a different way to investigate.

To draw the Bayesian awl

You need

1. To understand the (theoretical) estimand: what are you trying to do? Often not that easy to tell what a scientific study is trying to do. No vague research description.
2. A scientific (causal) model; estimand in 1. built in this context. We may resort to simulated data to convince ourselves.
3. To build a statistical models using 1. & 2. above. Can get out the estimand. Sometimes we cannot extract the estimand, but at least we will know we need a different way to investigate.
4. Simulate from 2. (in different ways) to validate (check) 3.

To draw the Bayesian awl

You need

1. To understand the (theoretical) estimand: what are you trying to do? Often not that easy to tell what a scientific study is trying to do. No vague research description.
2. A scientific (causal) model; estimand in 1. built in this context. We may resort to simulated data to convince ourselves.
3. To build a statistical models using 1. & 2. above. Can get out the estimand. Sometimes we cannot extract the estimand, but at least we will know we need a different way to investigate.
4. Simulate from 2. (in different ways) to validate (check) 3.
5. Analyse real data since our statistical model is expected to provide a satisfactory answer

- If any step above goes wrong, we can revise them and improve them: feedback.

Is this Saturn?

Or: Is is a statistical problem?



Is this Saturn?

Or: Is is a statistical problem?



- Bayesian approach is permissive and flexible.
- It expresses uncertainty at all levels.
- Bonus: direct solution to measurement errors or missing observations
- Focus on scientific modelling

MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Golem, owl and now DAGs



Golem, owl and now DAGs



Science comes before statistics

- For statistical models to produce scientific insight, the require additional/prior scientific (causal) models that contain how entities influence each other.
- The reasons for a statistical analysis is **not** found in the data themselves, but rather in the causes of the data. Data tables contain associations that can only be interpreted as causal in light of a (hypothetised) scientific causal model.
- Causes of the date cannot be extracted from the data alone: Not causes in, no causes out. Even description (describing the population from samples requires causal thinking) and experimental design are part of this. We will see example of possible biases

What is causal inference?

- Attempt to understand the scientific causal model.
- Two complementary components: (i) causal inference is prediction of interventions and (ii) causal inference is imputation of missing observations.
  These two are the same.

Causal inference as prediction

- Knowing the cause means being able to predict the consequences of an intervention: *What if I do this?*
- Example: wind blowing the trees. Which causes which? Both movements are always associated. If you can intervene and shake the trees, it won't create wind.

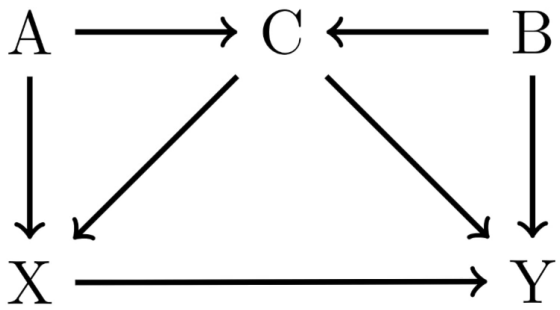
Causal inference is imputation

- Knowing the cause means being able to predict the consequences of an intervention: *What if I do this?*
- Example: wind blowing the trees. Which causes which? Both movements are always associated. If you can intervene and shake the trees, it won't create wind.

- Knowing a cause means being able to construct unobserved counterfactual outcomes (things that didn't happen): *What if I had done something else?*
- How would history had changed if China had been to the moon before the US? If you understand the mechanisms of history, you can predict what would have happened.

DAGs

- Directed Acyclic Graphs (see later)
- Heuristic causal models
- Clarify Scientific thinking
- Analyse logically to deduce appropriate statistical models



- Each letter is a variable (exists) and arrows represent causal relationship.
- Estimand: effect of X on Y. Other kind of things are going on. Different queries $\leftrightarrow$ different models.
- Question: which of A, B and/or C should you add to precisely estimate the effect $X \rightarrow Y$. Notion of control variable, confound. It's not safe to add everything.
- How to test a causal model? Testable implications? More scientific knowledge allows to do more, e.g. dynamical models!

MASSEY UNIVERSITY
TE KUNENGA KI PŪREHUROA
UNIVERSITY OF NEW ZEALAND

Summary

- Need to rethink popular statistical and scientific approaches.
- Build your golems (models) and know their strengths and limits.
- Bayesian inference built in steps (like drawing an owl) will be our way to go.
- DAGs: transparent scientific assumptions: justify scientific effort, expose it to critique, connect theories to golems.
- Next two weeks: really doing Bayesian inference.